



(An Autonomous Institution affiliated to VTU, Belagavi)

Supplementary Examinations - December 2025

21EEE143 / 20EE604 - VLSI CIRCUITS AND DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Unit – I		Marks	BT*	CO*	PO*
1	a) Describe the key steps involved in the VLSI design flow. Briefly explain each step and their significance in the overall design process.	10	L*2	1	1
	b) With the aid of neat sketch, discuss three regions of operation of an NMOs transistor. Discuss the condition under which each region occurs and the behavior of the transistor in each region.	10	L3	2	1,2,5,12
2.	a) Explain the different components of threshold voltage and hence calculate threshold voltage of an NMOs transistor with substrate grounded. The substrate doping density $N_A = 2 \times 10^{15}/\text{cm}^3$, polysilicon gate doping density of $N_D = 1 \times 10^{20}/\text{cm}^3$, gate oxide thickness of $t_{ox} = 1000 \text{ \AA}$ and oxide interface fixed charge density $N_{ox} = 1 \times 10^{10}/\text{cm}^2$. Assume no threshold adjust implantation impurities are added.	10	L3	2	1,2,5,12
	b) Explain the various fabrication steps involved in detail in hierarchical manner for an n-well CMOS process.	10	L2	1	1,2,5,12
3.	a) Describe the phenomenon of channel length modulation in NMOS transistors. Also explain how does this effect influence the transistor characteristics.	10	L3	2	1,2,5,12
	b) Compare the following. i) Enhancement and depletion Mos transistor ii) Mos transistor with BJT.	10	L3	1	1

Unit – II

4.	a)	Derive the effect of constant field and constant voltage scaling on MOSFET oxide capacitance, drain current, power dissipation, power density, and Short Gate area.	10	L3	3	1,2,5,12
	b)	Derive the expression for critical voltages V_{OH} and V_{OL} for a resistive load inverter. Hence solve. Consider a resistive load inverter circuit with $V_{DD} = 5V$, $K_n^1 = 20\mu A/V^2$, $V_{to} = 0.8V$, $R_L = 200k\Omega$ and $w/L=2$. Calculate the critical voltages (V_{OL} , V_{Ou}) on the VTC and find the noise margins of the circuit.	10	L3	3	1,2,5,12
5.	a)	What do you understand by stick diagram in VLSI design? State the rules to draw the stick diagram. Hence draw the stick diagram to realise $y = \overline{A + BC}$.	10	L3	3	1,2,5,12
	b)	With the help of diagram, explain the latch up problem in CMOS inverter.	10	L3	4	1,2,5,12
6.	a)	Explain the narrow channel effects in MOSFETS.	10	L2	3	1,2,5,12
	b)	Describe the voltage transfer characteristics (VTC) of a CMOS inverter, highlighting the operating regions of the NMOS and PMOS transistors. What factors influence the shape and slope of the VTC?	10	L3	4	1,2,5,12

P.T.O.

Unit – III

- | | | | | | |
|-------|--|----|----|---|----------|
| 7. a) | Derive an expression for critical voltages V_{OH} , V_{OL} of a two input nMOS NOR gate. | 10 | L3 | 5 | 1,2,5,12 |
| b) | Explain the following other form of CMOS logic circuits | | | | |
| | i) pseudo nMOS logic | | | | |
| | ii) Multiplexer | 10 | L2 | 5 | 1,2,5,12 |
| 8. a) | Explain the principle of operation of dynamic CMOS logic and discuss its key advantages and disadvantages compared to static CMOS logic. | 10 | L3 | 5 | 1,2,5,12 |
| b) | Design 2 input CMOS based NAND circuit and explain its operating. Also provide the stick and layout diagram layout for the same. | 10 | L4 | 5 | 1,2,5,12 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
